

Exercise 1.

- 5 culprits, 2 did it

Send 3 to detention?

- number of combinations of 3: $\binom{5}{3} = \frac{5!}{3!2!} = \frac{120}{12} = 10$
- favorable outcome: 2 guilty, 1 innocent

$$\binom{3}{1} = \frac{3!}{2!1!} = \frac{6}{2} = 3$$

$$\frac{\text{number of favorable outcomes}}{\text{number of outcomes}} = \frac{3}{10} = 0.3$$

Send 2 to detention?

- number of combinations of 2: $\binom{5}{2} = \frac{5!}{2!3!} = \frac{120}{12} = 10$
- favorable outcome: 2 guilty – > only 1 favorable outcome: $\binom{2}{2} = 1$

$$\frac{\text{number of favorable outcomes}}{\text{number of outcomes}} = \frac{1}{10} = 0.1$$

Send 4 to detention?

- number of combinations of 4: $\binom{5}{4} = \frac{5!}{4!1!} = \frac{120}{24} = 5$
- favorable outcomes 2 guilty, 2 innocent

At least 1 has to be guilty since there is only 1 child left that isn't chosen and 2 trouble makers.

$$\binom{3}{1} = \frac{3!}{1!2!} = \frac{6}{2} = 3$$

$$\frac{\text{number of favorable outcomes}}{\text{number of outcomes}} = \frac{3}{5} = 0.6$$

Exercise 2.

- 4 packages and 10 different houses
- each package has an equal chance of belonging to each house
- $|\Omega| = 10^4 = 10,000$
- number of combinations a package goes to 1 house: $10 \times 9 \times 8 \times 7 = 5,040$
 probability packages goes to 1 house = $\frac{5,040}{10,000} = 0.504$
 probability that a house receives more than 1 package = $1 - 0.504 = 0.496$

Exercise 3.

- number of combos of letters to choose: $\binom{11}{5} = \frac{11!}{5!6!} = 462$
- MISSISSIPPI: 4 vowels and 11 letters
- number of combos of 2 vowels: $\binom{4}{2}$
- number of combos of 3 consonants: $\binom{7}{3}$
- number of favorable outcomes: $\binom{4}{2} \times \binom{7}{3} = \frac{4!}{2!2!} \times \frac{7!}{3!4!} = 6 \times 35 = 210$

probability exactly 2 vowels selected: $\frac{210}{462} = \frac{5}{11}$

Exercise 4.

- number of ways to distribute 4 distinct aces to 4 people: $4! = 24$
- number of ways to distribute 4 distinct kings to 4 people: $4! = 24$
- number of ways to distribute the remaining 44 cards into 4 hands of 11 hands each:
 $\binom{44}{11, 11, 11, 11}$
- total number of ways to distribute 4 distinct aces and kings: $4! \times 4! \binom{44}{11, 11, 11, 11} =$
 $24 \times 24 \binom{44}{11, 11, 11, 11} = 576 \binom{44}{11, 11, 11, 11}$

Exercise 5.

- 12 distinct books
- boxes hold 3, 4, and 5 books

$$\binom{12}{3, 4, 5} = \frac{12!}{3!4!5!} = \frac{479,001,600}{17,280} = 27,720$$

4 books each?

- the boxes are now indistinguishable

$$\binom{12}{4, 4, 4} \times \frac{1}{3!} = \frac{12!}{4!4!4!3!} = \frac{12!}{(4!)^3 3!} = \frac{479,001,600}{82,944} = 5,775$$

Exercise 6.

- 16 people

- 4 teams (not distinct, order doesn't matter)
- 4 on each team

$$(a) \binom{16}{4, 4, 4, 4} \times \frac{1}{4!} = \frac{16!}{4!4!4!4!} \frac{1}{24} = \frac{16!}{(4!)^4} \frac{1}{24} = 2,627,625$$

(b)

- ways of distributing players from high school 1: $4! = 24$
- ways of distributing players from high school 2: $4! = 24$
- ways of distributing players from high school 3: $4! = 24$
- ways of distributing players from high school 4: $4! = 24$
- teams indistinguishable

$$(4!)^4 \times \frac{1}{4!} = (24)^4 \times \frac{1}{24} = 13,824$$

Exercise 7.

Probability that the sequence is descending

- Balls 1 - 9
- 4 are selected without replacement
- $|\Omega| = (9)_5 = 9 \times 8 \times 7 \times 6 = 3,024$

$$\frac{(9)_5}{4} = \binom{9}{4}$$

Therefore,

$$\begin{aligned} &= \frac{\binom{9}{4}}{3,024} \\ &= \frac{\frac{9!}{4!5!}}{3,024} \\ &= \frac{126}{3,024} \\ &= \frac{1}{24} \end{aligned}$$

Probability that the sequence follows the "up-down-up" pattern

- 1st number is smaller than the 2nd
- 2nd number is smaller than the 3rd
- 3rd number is smaller than the 4th

Position 1 and 3 can be swapped and 2 and 4 can be swapped and the pattern still holds. Also a 5th pattern can be created where the number in positions 1 and 2 can be larger than 3 and 4. So far every string of numbers there is $2!2! + 1 = 5$ combinations where the pattern holds true.

ex. With the numbers 1, 2, 3, and 4, 5 patterns can be made: 1, 3, 2, 4, 2, 3, 1, 4, 1, 4, 2, 3, 2, 4, 1, 3, and 3, 4, 1, 2

$$= 5 \times \binom{9}{4} \frac{1}{(9)_5}$$

$$= 5 \times \frac{126}{3,024}$$

$$= \frac{5}{24}$$

Exercise 8.

$$2^m = \sum_{j=0}^m \binom{m}{j}$$

$$j = k - 1$$

$$k = j + 1$$

Therefore,

$$m = n - 1$$

$$n = m + 1$$

$$2^{n-1} = \sum_{k=1}^n \binom{n-1}{k-1}$$

$$n2^{n-1} = n \sum_{k=1}^n \binom{n-1}{k-1}$$

$$n2^{n-1} = \sum_{k=1}^n n \binom{n-1}{k-1}$$

$$n2^{n-1} = \sum_{k=1}^n n \times \frac{(n-1)!}{(k-1)!(n-k)!}$$

$$n2^{n-1} = \sum_{k=1}^n \frac{n!}{(k-1)!(n-k)!}$$

$$n2^{n-1} = \sum_{k=1}^n \frac{k}{k} \times \frac{n!}{(k-1)!(n-k)!}$$

$$n2^{n-1} = \sum_{k=1}^n k \times \frac{n!}{k!(n-k)!}$$

$$n2^{n-1} = \sum_{k=1}^n k \binom{n}{k}$$

Exercise 9.

- 4 entry level job openings
- 37 female
- 7 males
- Total number of candidates: $37 + 7 = 44$
- chance of a candidate be chosen: $\binom{44}{4} = \frac{44!}{4!40!} = 135,751$
- chance 1 female is chosen out of 4: $\binom{37}{1} \times \binom{7}{3} = \frac{37!}{36!1!} \times \frac{7!}{3!4!} = 37 \times 35 = 1,295$
- chance 0 female is chosen out of 4: $\binom{7}{4} = \frac{7!}{4!3!} = 35$

At most 1 female is hired = $\frac{1,295+35}{135,751} = \frac{1,330}{135,751}$

Exercise 10.

- 12 long strings
- entries can be 1, 2, 3, 4
- $4^{12} = 16,777,216$ sample points equally likely

$$(a) \binom{12}{3,3,3,3} = \frac{12!}{(3!)^4} = 369,600$$

$$\frac{369,600}{16,777,216} = 0.0220$$

(b)

- 3 1's
- 9 of 2, 3, and 4s

$$3^9 = 19,683$$

- orders of these combos: $\binom{12}{3,9} = \frac{12!}{3!9!} = 220$

Therefore,

$$19,683 \times 220 = 4,330,260$$

$$\text{probability of exactly 3 1's} = \frac{4,330,260}{16,777,216} = 0.2581$$

(c)

- 3 1's
- 3 2's

- 6 3 and 4s

$$2^6 = 64$$

- orders of these combos: $\binom{12}{3,3,6} = \frac{12!}{3!3!6!} = 18,480$

$$64 \times 18,480 = 1,182,720$$

probability of exactly 3 1's and 3 2's = $\frac{1,182,720}{16,777,216} = 0.0705$

Exercise 11.

1. a, b, c, d, e, f are going to be assigned 3 distinct jobs
2. each person 1 job
3. job 1 = 3 people
4. job 2 = 2 people
5. job 3 = 1 person

(a) $\binom{6}{3,2,1} = \frac{6!}{3!2!1!} = \frac{720}{12} = 60$

(b) $\frac{1}{2} \times 60 = 30$

Exercise 12.

- 8 marbles 1-8 distinguishable
- 1, 2 are red
- 3, 4 are blue
- 5, 6 are green
- 7, 8 are yellow

(a) $4^8 = 65,536$

(b) $\binom{8}{2,2,2,2} = \frac{8!}{(2!)^4} = \frac{40,320}{16} = 2,520$

(c) $\frac{8!}{(2!)^4 4!} = \frac{40,320}{16 \times 24} = \frac{40,320}{384} = 105$

(d) $4! = 24$

(e) $\binom{4}{2} \times \binom{4}{2} = 6 \times 6 = 36$

Exercise 13.

$$(a + b + c)^n = \sum_{n_1+n_2+n_3=n, n_i \geq 0} \binom{n}{n_1, n_2, n_3} a^{n_1} b^{n_2} c^{n_3}$$

$$(-1 + 2x - y)^5$$

$$a = -1$$

$$b = 2x$$

$$c = -y$$

$$n = 5$$

$$(-1 + 2x - y)^5 = \sum_{n_1+n_2+n_3=5, n_i \geq 0} \binom{5}{n_1, n_2, n_3} (-1)^{n_1} (2x)^{n_2} (-y)^{n_3}$$

$$n_1 + n_2 + n_3 = 5$$

1. $5 + 0 + 0 = 5$
2. $4 + 1 + 0 = 5$
3. $4 + 0 + 1 = 5$
4. $3 + 1 + 1 = 5$
5. $3 + 2 + 0 = 5$
6. $3 + 0 + 2 = 5$
7. $2 + 3 + 0 = 5$
8. $2 + 0 + 3 = 5$
9. $2 + 2 + 1 = 5$
10. $2 + 1 + 2 = 5$
11. $1 + 4 + 0 = 5$
12. $1 + 0 + 4 = 5$
13. $1 + 3 + 1 = 5$
14. $1 + 1 + 3 = 5$
15. $1 + 2 + 2 = 5$
16. $0 + 5 + 0 = 5$
17. $0 + 0 + 5 = 5$
18. $0 + 4 + 1 = 5$
19. $0 + 1 + 4 = 5$

$$20. 0 + 3 + 2 = 5 **$$

$$21. 0 + 2 + 3 = 5$$

$$n_1 = 0$$

$$n_2 = 3$$

$$n_3 = 2$$

$$= \binom{5}{0, 2, 3} (-1)^0 (2x)^3 (-y)^2$$

$$= \binom{5}{0, 2, 3} 1 \times 8x^3 \times y^2$$

$$= \frac{5!}{0!2!3!} \times 8x^3 y^2$$

$$= \frac{120}{1 \times 2 \times 6} \times 8x^3 y^2$$

$$= \frac{120}{10} \times 8x^3 y^2$$

$$= 10 \times 8x^3 y^2$$

$$= 80x^3 y^2$$

COEFFICIENT: 80